

AKSHAY SASI

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OVERVIEW

AI/ML Engineer specializing in Machine Learning, Deep Learning, and LLM-based systems, with experience in building and deploying production-grade AI applications, including multi-agent systems, RAG pipelines, LLM guardrails, and scalable backend services. Key Skills: Python, Machine Learning, Deep Learning, NLP, LLMs, RAG, FastAPI, PyTorch, Docker, AWS

EXPERIENCE

- AI & ML Engineer** June 2025 - Present
Storygame (P) Ltd Trivandrum, Kerala
 - Architected multi-agent LLM backends using LangChain, LangGraph and MCP integrated automation pipelines with tool-calling, memory handling, and task execution across complex agentic workflows.
 - Conducted LLM fine-tuning on virtual GPU infrastructure and implemented voice-agent systems integrating speech-to-text with LLM reasoning for optimized performance and cost-efficient inference.
- AI Intern** Jan 2025 - June 2025
Quest Global Trivandrum, Kerala
 - Researched the latest AI/ML and NLP frameworks for clustering and predictive modeling.
 - Developed predictive and clustering models, improving performance through HDBSCAN and feature engineering.
- Research & Development Intern** Apr 2024 - July 2024
AI Chip Center - Digital University Kerala Hybrid - Trivandrum
 - Designed custom compilers and processor architectures for chips using Verilog and X-HEEP.
 - Optimized RISC-V architecture, reducing inference latency by 15% for ML workloads.

EDUCATION

- M.Tech, Computer Science and Engineering (AI Specialization)** Sep 2023 - May 2025
Digital University Kerala, Trivandrum, Kerala Trivandrum, Kerala
 - CGPA: 8.0/10
 - Relevant Coursework: Mathematics in ML & DL, Advanced data structures, Deep learning, NLP, Cloud computing, Cognitive computing and Computer vision.
- B.Tech, Computer Science and Engineering** Aug 2019 - Aug 2023
Vimal Jyothi Engineering College (KTU), Chemperi, Kerala Kannur, Kerala
 - CGPA: 7.64/10
 - Relevant Coursework: Data structures, OOPS, OS, Computer architecture, Microcomputers, Database management, Artificial Intelligence, Graph Theory and Networking.

SELECTED RESEARCH PROJECTS

- GROOVE: A Generative Refinement Framework Using Vision-Language Feedback Loops** June 2025
Tools: Python, Transformers, Diffusion Models, Vision-Language Models 
 - Developed an agentic system to bridge the semantic gap in text-to-image synthesis by autonomously refining prompts to reproduce target visual concepts.
- Unveiling the Heart-Brain Connection: An Analysis of HRV in Cognitive Performance (Thesis)** May 2025
Tools: Python, Neurokit2, MNE, ML/DL Algorithms 
 - Created a multimodal cognitive load classifier using ECG (HRV, Catch22) and EEG, achieving 92% accuracy for non-invasive mental state monitoring.
- Texture-Based Food Contamination Detection Framework in Agricultural Products (Term paper)** Dec 2024
Tools: Python, PyTorch, OpenCV
 - Designed a PyTorch-based computer vision pipeline leveraging texture analysis and morphological component extraction for automated contamination detection.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

[S.1] Akshay Sasi, Malavika Pradeep, Nusaibah Farrukh, Rahul Venugopal, and Elizabeth Sherly (2026). **Unveiling the Heart-Brain Connection: An Analysis of ECG in Cognitive Performance**. *arXiv preprint*, arXiv:2601.01424.

AI SYSTEMS & PRODUCTS

- **Omnella - Open-Source Multi-Agent Marketing Automation Platform** May 2026
Tools: Python, Multi-Agent Systems, LLM APIs, SSE, Docker, Next.js 
 - Built an open-source multi-agent orchestration platform, custom agentic engine with tool use, memory, and retry logic driving four specialized agents across a provider-agnostic LLM layer, with real-time SSE streaming, Docker self-hosting, and zero server-side key exposure.
- **GlassLM – Privacy-Preserving LLM Middleware** Dec 2025
Tools: JavaScript, Python, Browser APIs, NER, LLM APIs Live | 
 - Architected an open-source LLM guardrails middleware with client-side prompt sanitization and reversible placeholder tokenization, ensuring zero sensitive data leaves the user’s local environment

SKILLS

- **Generative AI LLM Systems:** RAG pipelines, Prompt engineering, Multi-agent systems, LangGraph, Agentic workflows, MCP, LLMOps, SLM/LLM integration
- **Voice Multimodal AI:** ASR TTS integration, Voice agent pipelines, Lip-sync model inference, Image video generation workflows
- **Machine Learning:** Classification prediction models, PyTorch, HuggingFace Transformers, Scikit-learn, Model evaluation
- **Backend Deployment:** FastAPI, REST APIs, n8n workflow architecture, Docker, Kubernetes (basic), AWS (EC2, S3), Production deployment

HONORS AND ACHIEVEMENTS

- **Best Employee Award – QX 2025** Nov 2025
Storygame (P) Ltd
 - Recognized for contributions to building and deploying AI-driven systems, including agentic workflows and voice AI solutions.
- **Academic Excellence Award – M.Tech (AI)** Oct 2025
Digital University Kerala
 - Secured Second Rank in M.Tech with specialization in Artificial Intelligence.
- **1st Prize – ICT Day Hackathon** Apr 2025
Digital University Kerala
 - Built an AI-powered anonymous social platform using BERT and React Native.
- **3rd Prize – Ethical AI Ideathon** Sept 2024
Digital University Kerala & Mozilla Responsible Computing Challenge
 - Proposed an NLP-based content diversification system addressing filter bubble bias.

LEADERSHIP & ACTIVITIES

- **Project Lead – B.Tech Major Project** 2022 - 2023
Vimal Jyothi Engineering College
 - Led a 4-member team in developing a privacy-focused Android application with a face recognition module (95% accuracy).
- **Chief Creative Officer – Innovation Club** 2024
Digital University Kerala
 - Coordinated cross-functional student teams and managed 5+ innovation events.

CERTIFICATIONS

- **NPTEL Swayam:** Introduction to Large Language Models Apr 2025
- **NVIDIA Generative AI/LLM:** Building RAG Agents Jul 2025